

A Non-Intrusive Approach Monitors Patients with PIMD Using Environmental Sensors

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Introduction

Profound Intellectual and Multiple Disabilities (PIMD) patients are a population with severe intellectual and sensory impairments, requiring continuous monitoring from caregivers to ensure safety. While contact-based methods provide physiological information directly, they may cause discomfort in patients. To address this gap, we propose a non-intrusive, environment-based approach to monitor both patients and their surroundings without direct physical contact.

Background Information

- Limitations of Intrusive monitoring approaches (e.g., wearable sensors, on-body pads)
 - Discomfort or agitation in patients triggered by physical contact
 - High deployment and maintenance costs of wearable sensors
- Environmental sensors replacing on-body devices where possible
- Data pipeline: real-time inference performed by Jetson Orin Nano, model training conducted by backend server via daily patient data transmission

Objectives

- Harness useful signals and information from surroundings, for example
 - Patient utterances isolated from background noises
 - rapid shifts in environmental conditions
- A base model that detects anomalous or noteworthy events and can be fine-tuned for individual patients
- Real-time performance within Jetson Orin Nano's 8GB memory constraint

Methodology

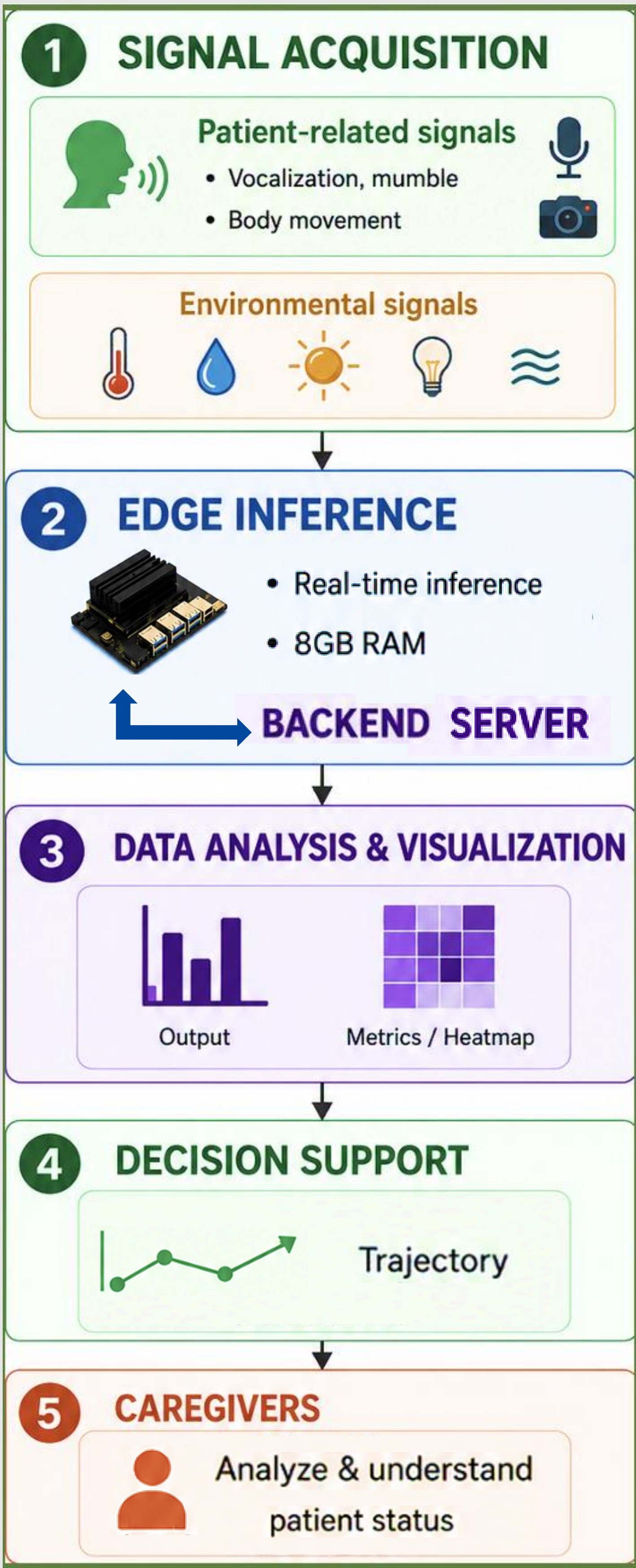


Figure 1. Voice4PIMD five-stage sensing and monitoring pipeline.

1. Multi-stream signal acquisition: patient-related signals captured alongside environmental signals
2. Real-time edge inference performed by an NVIDIA Jetson Orin Nano, constrained to 8GB unified memory
 - Constraint requires a lightweight model footprint, trading some complexity for memory
3. Detection-oriented rather than classification-oriented model design, since vocalization meanings vary substantially across individual patients
 - Model flags anomalous or noteworthy events rather than interpreting them, deferring interpretation to caregivers to enable generalization across a diverse patient population
4. Offline model training on a backend server using daily raw data uploads
5. Structured, time-stamped event logs aggregated into caregiver-facing metrics, heatmap summaries, and a longitudinal patient trajectory
 - Supporting both immediate awareness of flagged events and trend-based decision-making over time

Results and Future Projections

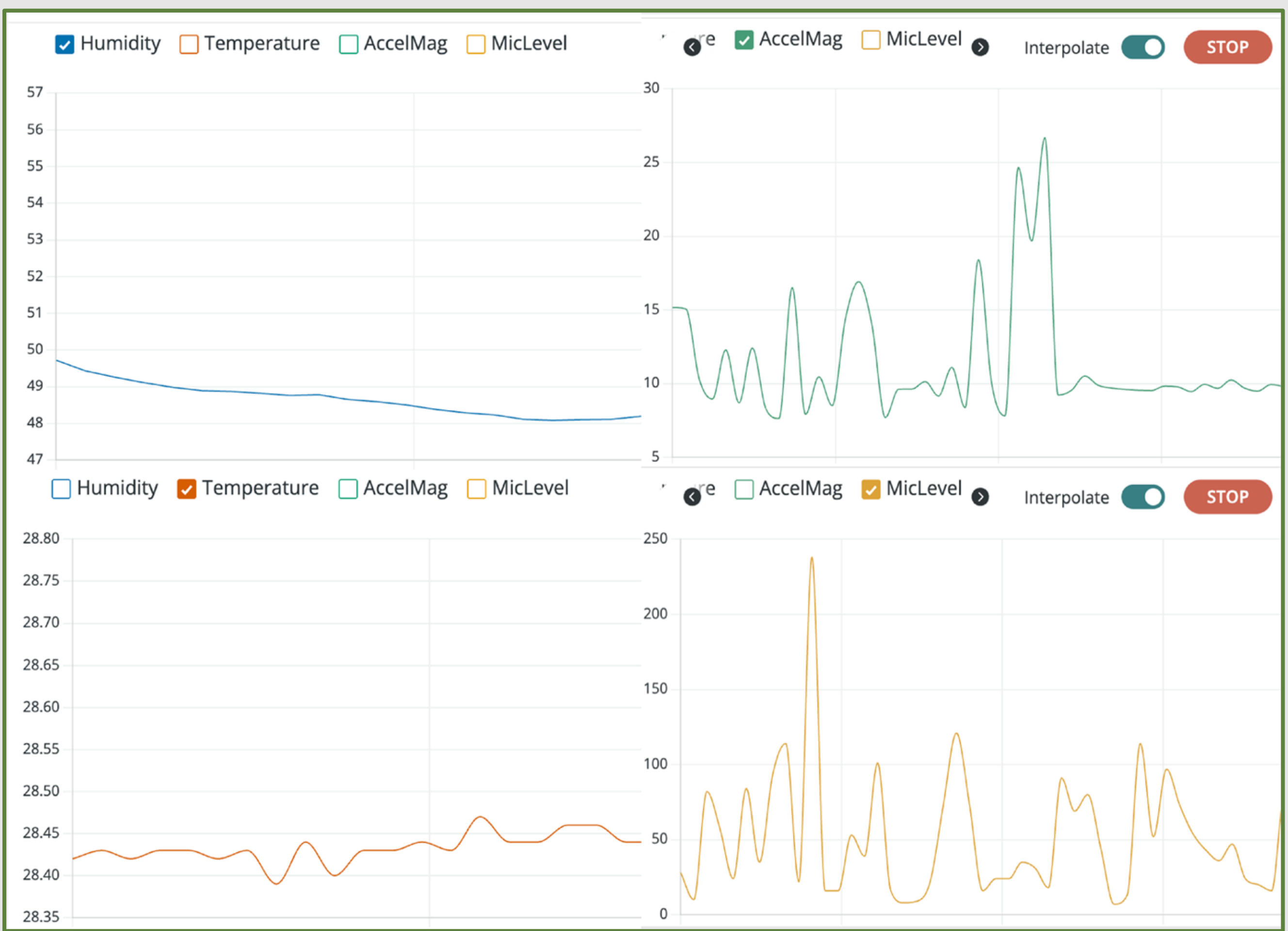


Figure 2. Real-time sensor readings from Feather Sense demo board.

Findings:

- As a preliminary demonstration, Adafruit Feather Sense board nRF52840 is used to validate the feasibility of the proposed sensing pipeline
- The accelerometer and microphone showed strong sensitivity to movements and sound events, while the humidity and temperature sensors maintained stable with high resolution

Future projections :

- Expand sensing modalities to enable richer and more comprehensive monitoring
 - Develop a data analysis method to extract meaningful information from collected multimodal signals, with potential integration to support better model learning
- Identify and leverage publicly available datasets or comparable scenarios to further train and evaluate AI models
 - Extend beyond the PIMD context, drawing on broader real-world audio sources such as film dialogue that contains comparable mixes of speech and background noises
 - Benefit from fewer ethical and licensing constraints than clinical data collection

References :

G. Slapničar et al., “Intelligent Supportive System for People with Profound Intellectual and Multiple Disabilities,” Advanced Intelligent Systems, vol. 8, Art. 2400925, 2026. doi:10.1002/aisy.202400925.

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